Aabhash Rahut

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Professional Summary

AI/ML-focused Computer Science undergraduate skilled in Python, SQL, Machine Learning, Deep Learning, Computer Vision, and Full-Stack development. Experienced in building Flask-based AI applications, recommendation systems, image classification models, and optimized PyTorch workflows for Apple Silicon.

Technical Skills

Languages	Python, SQL
Frameworks & Libraries	Pandas, NumPy, Scikit-Learn, Matplotlib, TensorFlow, Streamlit, Flask
Tools	MS Word, PowerPoint, MySQL
Platforms	Jupyter Notebook, Visual Studio Code, PyCharm
Soft Skills	Documentation Work, Rapport Building, Excellent Communication

Projects

- **Divya Drishti - Visually Impaired AI Assistant** REPO
  Flask, Python, OpenCV, TensorFlow, Text-to-Speech *March 2026*

Developed an AI-powered assistive application using Flask, Python, OpenCV, and TensorFlow, enabling visually impaired users to identify objects and receive real-time audio guidance. Integrated computer vision, image preprocessing, and Text-to-Speech (TTS) technologies to perform object detection and text recognition, enhancing accessibility and user interaction.

- **Restaurant Recommendation Model** REPO
  Flask, HTML, CSS, JavaScript, Pandas, NLTK *November 2025*

Developed a full-stack Flask web application with an interactive frontend using HTML, CSS, and JavaScript, enabling users to filter recommendations by cost range, minimum rating, and favourite restaurants. Performed end-to-end data preprocessing and feature engineering on a Zomato dataset of Bangalore using Pandas and NLTK, creating optimized features to improve model relevance and user filtering.

- **Plant Disease Prediction Model** REPO
  TensorFlow, Keras, CNN, Image Classification *September 2025*

Developed a deep learning-based image classification system using a Convolutional Neural Network (CNN) in TensorFlow/Keras to accurately detect and classify 38 different categories of plant diseases from leaf images. Achieved 99.21% training accuracy and 96.86% validation accuracy, implementing an 80/20 train-validation split and preparing a separate test set to ensure rigorous model evaluation.

Research

- **MobileNetV4 Optimization Apple Silicon** REPO
  PyTorch, MobileNetV4, CIFAR-10, Apple Silicon MPS *October 2025*

Optimized a PyTorch implementation of MobileNetV4, achieving a 13.19% absolute accuracy boost from 70.09% to 83.28% on CIFAR-10 by integrating Squeeze-and-Excitation blocks, Stochastic Depth regularization, and SiLU activation functions. Accelerated training time by 3.3x through strategic optimization for Apple Silicon GPUs via PyTorch's MPS backend.

Education

VIT Bhopal University, Sehore, MP — Sept 2023 – Jan 2027
Bachelor of Technology – Computer Science & Engineering (AI/ML) — **GPA: 8.19**

Griffins International School (GIS), Kharagpur, WB — April 2023
Senior School Certification Examination (Class 12 Boards) — **Score: 87.6%**

Delhi Public School (DPSJ), Jamnagar, Gujarat — April 2021
Secondary School Certification Examination (Class 10 Boards) — **Score: 94%**

Certifications

Microsoft Azure Data Fundamentals (DP-900) — **Score: 940/1000**
Microsoft. Covered Azure data services, relational and non-relational databases, OLTP/OLAP workloads, Azure SQL Database, Cosmos DB, and Synapse Analytics.

Cloud Computing – NPTEL (IIT Kharagpur) — **Score: 77%**
Elite + Silver Certification. Studied cloud service models, deployment models, virtualization, containerization, cloud storage, databases, security, and cost optimization.

Introduction to Internet of Things (IoT) – NPTEL (IIT Kharagpur) — **Score: 88%**
Elite + Silver Certification — Top 5% of certified candidates. Covered IoT architecture, sensors, actuators, embedded systems, communication protocols, cloud-enabled IoT applications, data acquisition, and smart system design.